Cloud Computing Practicals 5–10: Installation Steps

Subject: Cloud Computing | **Course:** M.Sc. IT Part I, Semester I

This document collects the installation steps for Practicals 5 to 10. Each practical lists its aim, requirements, steps and the screenshots to paste as output.

Practical 5: Xen Virtualization with XenCenter

Aim: Implement Xen virtualization and manage it with XenCenter.

Requirements: VMware Workstation, XenServer ISO, Citrix XenCenter (Windows), Windows XP SP3 ISO shared from a folder on your PC, VT-x/AMD-V enabled in BIOS, 8 GB or more RAM.

Steps:

1. In VMware Workstation choose **File → New Virtual Machine → Custom**, select the XenServer ISO and set the guest OS to **Linux**.
2. Give the VM 2 or more processors, 4 GB or more RAM and a 60 GB disk. Set the network to Bridged or NAT and tick **Virtualize Intel VT-x/EPT or AMD-V/RVI**.
3. Power on and run the installer: keyboard layout, accept EULA, select the disk, choose **Local media**, set the **root password**, set the IP and hostname, then choose the time zone.
4. After the reboot, note the IP address on the console (for example 192.168.124.137) and ping it from your PC.
5. Install **Citrix XenCenter** on your PC if needed, open it and click **Add a Server**. Enter the IP, user `root` and your password, then click **Add**.
6. Click **New Storage**, choose **Windows File Sharing (CIFS)**, untick auto-generate, enter the share path of the folder holding the XP ISO plus your PC credentials, and click **Finish**.
7. Click **New VM**, select **Windows XP SP3**, choose the ISO, accept the defaults, untick **Start the new VM**, and click **Create Now**.
8. Right-click the VM, choose **Start**, open the **Console** and complete the setup. The Windows XP welcome screen means success.

Output: XenServer console with the IP, the ping reply, the server added in XenCenter, the CIFS storage, the VM wizard, and the XP welcome screen.

Practical 6: VMware ESXi with vCenter

Aim: Implement virtualization using VMware ESXi Server and manage it with vCenter / vSphere.

Requirements: VMware Workstation, ESXi ISO, Windows XP ISO, VT-x/AMD-V enabled in BIOS, 8 GB or more RAM.

Steps:

1. In VMware Workstation create a new VM from the ESXi ISO. Set the guest OS to **VMware ESX**, with 2 or more processors, 4 GB or more RAM and a 40 GB disk, and tick **Virtualize Intel VT-x/EPT or AMD-V/RVI**.
2. Power on, press **Enter**, accept the license with **F11**, select the disk, choose the keyboard layout, set the **root password**, and press **F11** to install. Reboot when done.
3. Note the IP shown on the yellow ESXi screen and ping it from your PC.
4. Open `https://<ESXi-IP>` in a browser (or the vSphere Client) and log in as `root`.
5. Upload the Windows XP ISO to the datastore under **Storage → datastore1 → Datastore browser → Upload**.
6. Choose **Virtual Machines → Create / Register VM**, set the guest OS to Windows XP, select the datastore, and set 1 CPU, 512 MB to 1 GB RAM and a 10 GB disk. Set the CD/DVD drive to **Datastore ISO file** with **Connect at power on** ticked.
7. Power on the VM, open the console and complete the XP installation.
8. Optional: deploy the **vCenter Server Appliance**, log in to the vSphere Client, choose **Create Datacenter → Add Host**, and add the ESXi host. VCSA needs about 12 GB or more of RAM.

Output: ESXi console with IP, the ping reply, vSphere login, host summary, VM wizard, and XP running.

Practical 7: Server Cluster with Windows Server 2012

Aim: Implement a server cluster with Windows Server 2012.

Requirements: Two Windows Server 2012 machines (Node1, Node2) in the same domain, one shared iSCSI target visible to both nodes, and static IP addresses on all machines.

Steps:

1. On the machine that will provide storage (a third server, or Node1), open **Server Manager → Add Roles and Features → File and Storage Services → iSCSI Target Server** and install it.
2. Open **File and Storage Services → iSCSI** and choose **New iSCSI Virtual Disk**. Give it a name and size, create a new target, and add both nodes as initiators (by IP or IQN).
3. On **both nodes**, open **Server Manager → Add Roles and Features → Features**, select **Failover Clustering**, and install it.
4. On both nodes, open **iSCSI Initiator** from Administrative Tools and enter the target IP. Use **Quick Connect**, then connect to the discovered target.
5. On Node1, open **Disk Management**, bring the new iSCSI disk online, initialise it, and format it with NTFS.
6. Open **Failover Cluster Manager** and choose **Validate Configuration**. Add Node1 and Node2 and fix any errors reported.
7. Choose **Create Cluster**, add both nodes, and enter the cluster name and a static cluster IP.
8. Under **Storage → Disks**, add the shared disk. Choose **Configure Role** and add a role such as **File Server**.
9. Test failover: right-click the role, choose **Move → Select Node**, and confirm it comes up on Node2.

Output: The validation report, the cluster summary, and the role before and after failover.

Practical 8: Virtualization using Hyper-V

Aim: Implement virtualization using Hyper-V.

Requirements: Windows 10 or 11 Pro/Enterprise (or Windows Server) with virtualization enabled in BIOS, and a Windows XP ISO.

Steps:

1. If VMware Workstation is installed, uninstall it first through **Control Panel → Uninstall a program**, because it conflicts with Hyper-V on older versions.
2. Click **Turn Windows features on or off**, tick **Hyper-V** (with its management tools and platform), and click **OK**. Restart when asked.
3. After the restart, search for **Hyper-V Manager** and open it.

4. Click **Virtual Switch Manager**, choose **External** as the connection type, click **Create Virtual Switch**, name it, select your physical network adapter, and click **OK**.
5. Choose **New → Virtual Machine**. Give it a name, choose a generation (Generation 1 suits XP), assign RAM, and connect it to the external switch.
6. Create a virtual hard disk, then choose **Install an operating system from a bootable CD/DVD → Image file (.iso)** and select the Windows XP ISO.
7. Click **Finish**, right-click the VM, choose **Connect**, and click **Start**. Complete the XP installation.

Output: Windows Features with Hyper-V ticked, the Virtual Switch Manager, the VM settings, and XP running.

Practical 9: IaaS with Eucalyptus

Aim: Implement IaaS with Eucalyptus.

Requirements: A clean CentOS/RHEL machine with virtualization enabled, a static IP and internet access, and a free IP range for instances.

Steps:

1. Set a static IP address, then update the system with `yum update -y`.
2. Disable conflicting services: stop and disable the firewall and set SELinux to permissive or disabled (`systemctl stop firewalld`, `systemctl disable firewalld`, `setenforce 0`).
3. Confirm the machine supports virtualization with `egrep -c '(vmx|svm)' /proc/cpuinfo` (the result should be 1 or more).
4. Download and run the **Eucalyptus FastStart** installer script from the Eucalyptus website as root.
5. Answer the prompts: network mode, the IP address of the machine, and the free public IP range for instances.
6. Wait for the installation to finish and note the **console URL** (usually `https://<server-IP>:8888`) and the login details printed by the installer.
7. Log in to the console, register or upload an image, create a key pair and security group, then choose **Launch Instance**.
8. Confirm the instance reaches the **running** state.

Output: The installer finishing, the console dashboard, and the running instance.

Practical 10: OpenNebula

Aim: Implement OpenNebula.

Requirements: A clean Ubuntu or CentOS machine with internet access, virtualization enabled, and sudo or root access.

Steps:

1. Update the system (`sudo apt update && sudo apt upgrade -y` on Ubuntu) and note the machine's IP address with `ip a`.
2. Download the **miniONE** installer from the OpenNebula GitHub releases page (for example `wget https://github.com/OpenNebula/minione/releases/latest/download/minione`).
3. Run the installer as root: `sudo bash minione`.
4. Wait for the installation to finish and note the **Sunstone URL**, the username (`oneadmin`) and the password printed at the end.
5. Open `http://<server-IP>:2616` in a browser and log in as `oneadmin` with that password.
6. Go to **Infrastructure → Hosts** to see the available hosts. You can also launch a VM from **Instances → VMs → Create**.

Output: The miniONE finish message, the Sunstone login, and the Hosts page.